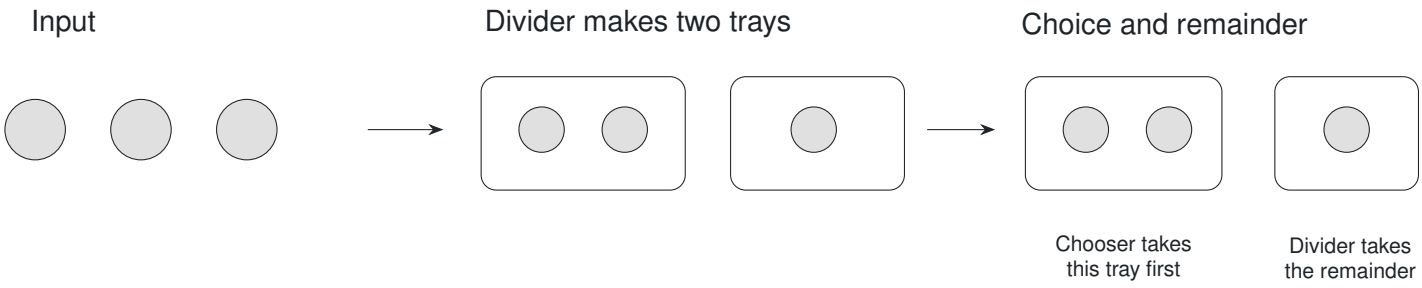
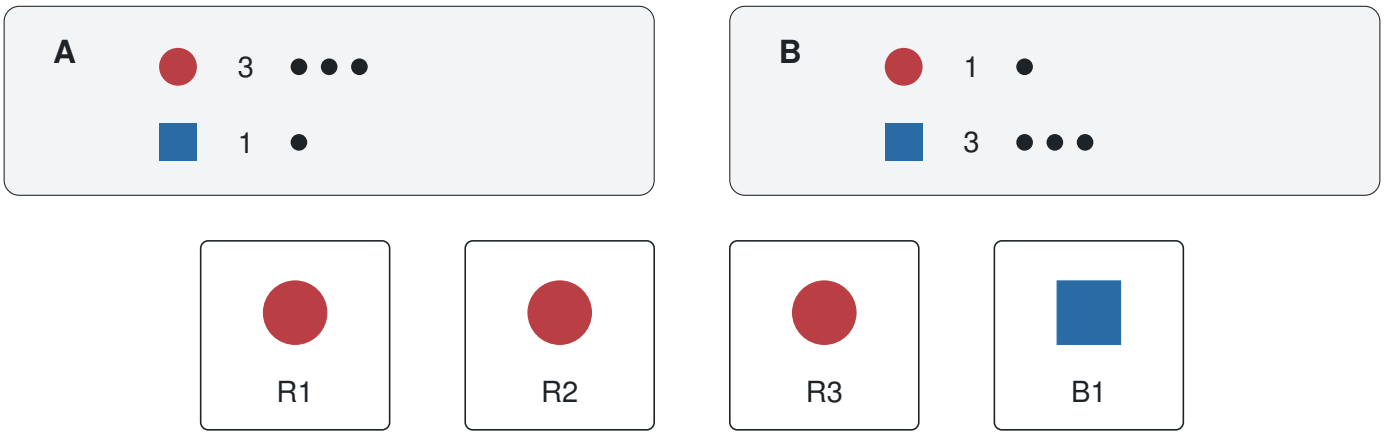
Keep each person's preference card unchanged during a trial. A tray's value is the sum of its cards' values, using the same person's preferences for every tray. Use every card once. Cards stay whole unless the problem allows cutting. Ties are allowed.



Problem 1: Use preference cards A and B. Divide these cards into two trays; your partner chooses a tray first, and you take the remainder. Find divisions where both people would keep their tray instead of trading. Swap who divides.



A	B

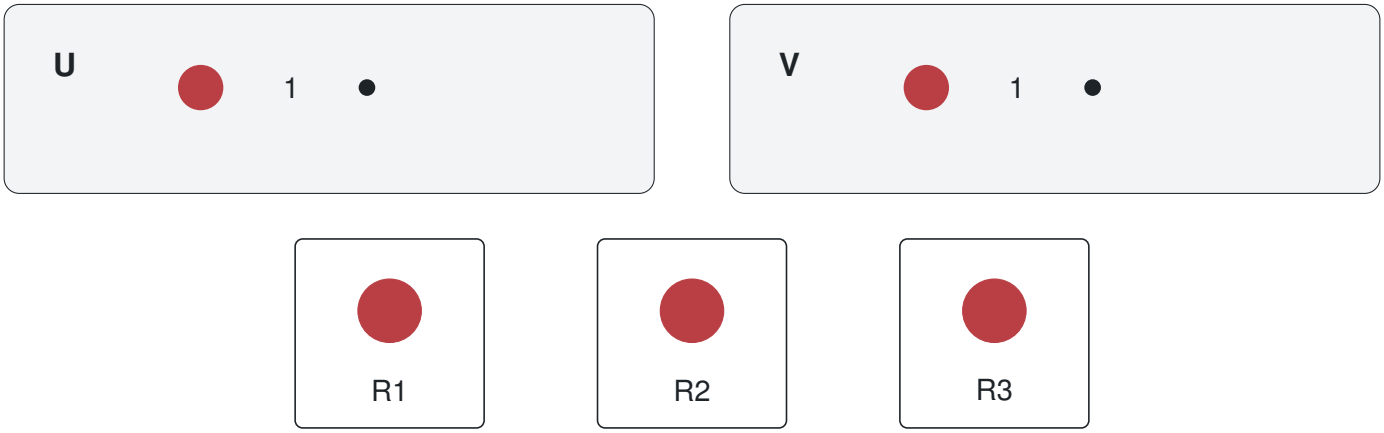
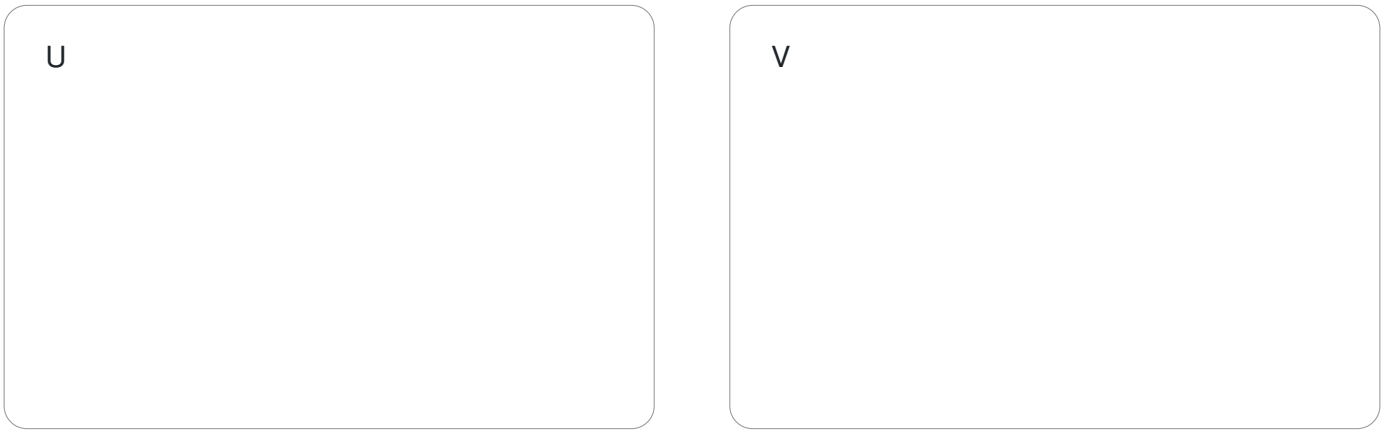
A	B

A	B

A	B

A	B

Problem 5: U and V each value every red card at 1. Divide all three cards between them, keeping each card whole, so neither person envies the other. If no division works, explain why. For a second trial, replace R3 with its paper copy; you may cut that copy. R1 and R2 must stay whole. Can you now make an envy-free division?



In the second trial, R3's whole paper square is worth 1 to each person. Equal areas have equal value.